Hyperparameter Tuning in Support Vector Machines: Investigating the Influence of C and Gamma

Student name: Sherly **Student id:** 25000121

GitHub: https://github.com/Shreya-291/Machine-Learning

Introduction

A common obstacle when evaluating machine learning models is pre-training parameter selection. The factors; hyperparameters that govern the basic working of the algorithm with respect to its data are called. Proper choice of hyperparameter values is required as they directly influence performance, complexity, and Generalization of the model.

Support Vector Machines (SVMs) are effective classification algorithms that have strong theoretical grounds with high predictive accuracy. The SVM method builds a decision surface that separates observations belonging to different classes. The main goal of SVM is to identify the maximum-margin separating hyperplane between the data points in the training set.

Although the underlying principle of SVM is relatively straightforward, the behaviour of the classifier depends heavily on the values of its hyperparameters. In particular, two parameters play a crucial role in determining how the model behaves during training:

- C (regularization parameter)
- Gamma (kernel coefficient)

The process of choosing values for these parameters is called hyperparameter tuning. Without appropriate tuning, a certain model will be too simple to catch important patterns (underfitting) or too complex and will memorize the training data (overfitting).

This tutorial explains how varying C and gamma values affect the Support Vector Machines behaviour. Using the Iris dataset, the demonstration of Decision Boundary Visualisation and Performance Comparison shows how these hyperparameters affect model complexity and classification performance.

In this tutorial, we will look at how hyperparameter tuning increases the SVM performance and also understand the parameters of an SVM which is necessary to build a model.

Fundamentals of Support Vector Machines

Support Vector Machines are algorithms that use supervised learning to classify data. In a classification problem, input features are passed to a learning algorithm and labelled training data is used to predict the correct class of the new observation.

The concept of SVM is to choose a decision boundary which separates the dataset into different classes. In mathematics, the boundary is called Hyperplane.

If the data has only two features, hyperplane would be straight line in two dimensions.

Support Vector Machine (SVM) is a type of classification algorithm that works by maximizing the distance between the decision boundary and the closest values from both classes. This gap is known as the margin.

Support vectors refer to the data points that are closest to the separating hyperplane. These points are important as they lead to the positioning of the decision boundary. Maximizing the margin enables the classifier to predict unseen data better.
Even so, many real datasets cannot be separated by a simple straight line. When the data is not linearly separable, kernel functions enable the model to create nonlinear decision boundaries that capture the data's structure more accurately.

Hyperparameters and Model Behaviour

Although SVM provides a powerful framework for classification, the behaviour of the algorithm is controlled by hyperparameters. Two hyperparameters are particularly important when using the Radial Basis Function (RBF) kernel: **C** and **gamma**.

- These parameters determine how the decision boundary adapts to the training data.
- Understanding their influence is essential for building effective SVM models.

The Role of the C Parameter:

The parameter C acts as a regularization parameter that determines how strongly the model penalizes classification errors during training.

When the value of C is small, the classifier allows some training observations to be misclassified in order to maintain a wider margin. This results in a simpler decision boundary that may generalize better to new data.

In contrast, when C is large, the algorithm attempts to classify every training example correctly. This forces the model to create a narrower margin and a more complex decision boundary.

As a result, the C parameter governs the trade-off between

- increasing the margin.
- Reducing classification mistakes.

If C is very small, then the model may underfit the data because the boundary is too simple. When C is a large numbered value, the model may overfit to the training data.

The Role of the Gamma Parameter

The gamma parameter specifies the extent to which a single training sample can reach in the process of kernel application, particularly with RBF kernel.

When the gamma value is small then every training point has a large region of influence. The decision boundary generated by this model is smooth and varies gradually.

A high gamma value confines the impact of each training data point to a limited area. As a result, the classifier yields a very detailed decision boundary that closely adheres to the training data.

If gamma is too high, then the model could become too sensitive to individual training samples and overfit.

Dataset Overview:

The dataset used in this experiment is the **Iris dataset**, which is one of the most commonly used datasets in machine learning education.

The dataset contains measurements of iris flowers belonging to three species:

- Iris setosa
- Iris versicolor
- Iris virginica

Each observation includes four numerical features describing the flower:

- sepal length
- sepal width
- petal length
- petal width

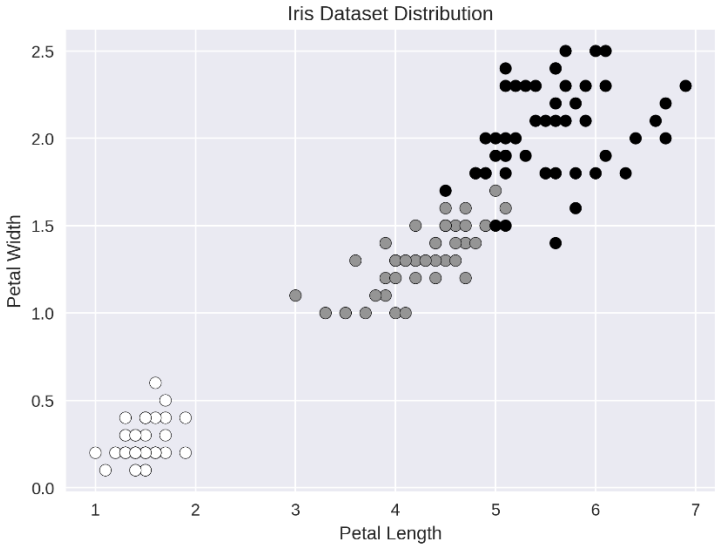
To visualize the decision boundaries clearly, two features were selected for this experiment:

- petal length
- petal width

Using two features allows the behaviour of the SVM model to be illustrated in two-dimensional space.

Data Exploration

Before training the model, the dataset was visualized in order to examine how the classes are distributed.



The visualization shows that the three species form identifiable clusters in the feature space. However, some overlap occurs between the versicolor and virginica classes, making the classification problem more challenging.

Model Training Procedure

The input features were scaled prior to training the Support Vector Machine models. Standardization refers to changing each feature so that it is rescaled to have a mean of zero and a standard deviation of one.
To determine the position of the decision boundary, SVM models rely on distance calculation for which Feature Scaling is necessary.
After performing feature scaling, the data set was divided into the training and test set. The model was built by using training data whereas testing data was used to evaluate the performance of the model.

Four SVM models were trained in order to analyze the impact of hyperparameters:

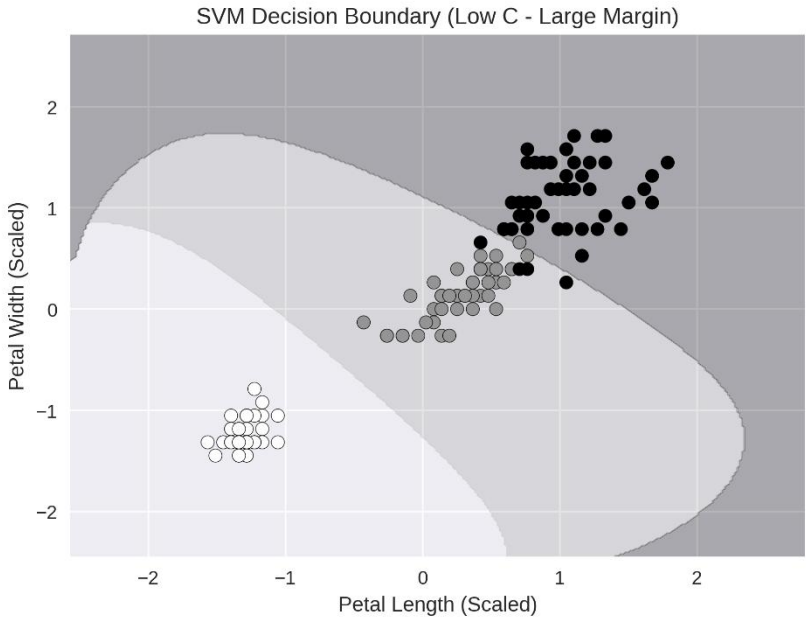
- SVM with a low C value
- SVM with a high C value
- SVM with a low gamma value

- SVM with a high gamma value

Influence of the C Parameter

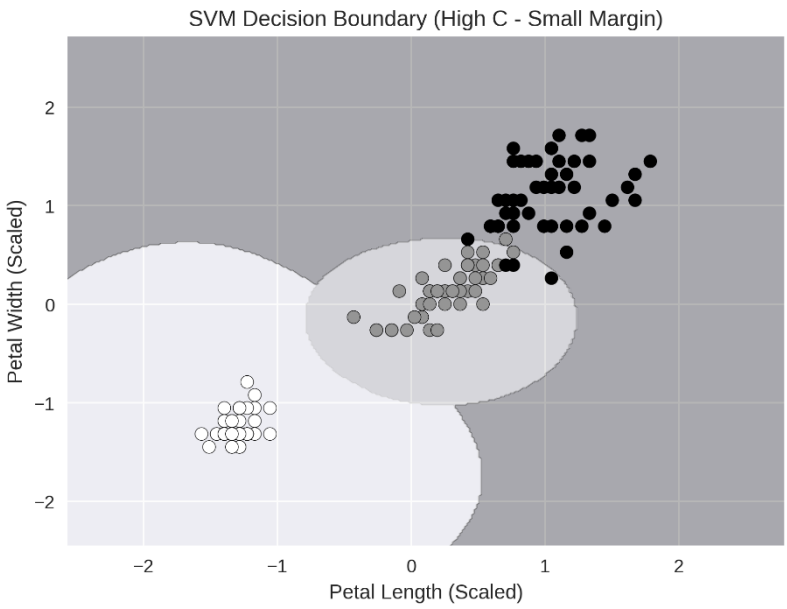
The effect of the C parameter can be observed by comparing the decision boundaries generated by models with different C values.

Decision boundary produced using a low C value



When the value of C is small, the classifier allows some misclassification in order to maintain a wider margin. This produces a smoother boundary that is less sensitive to individual data points.

Decision boundary produced using a high C value

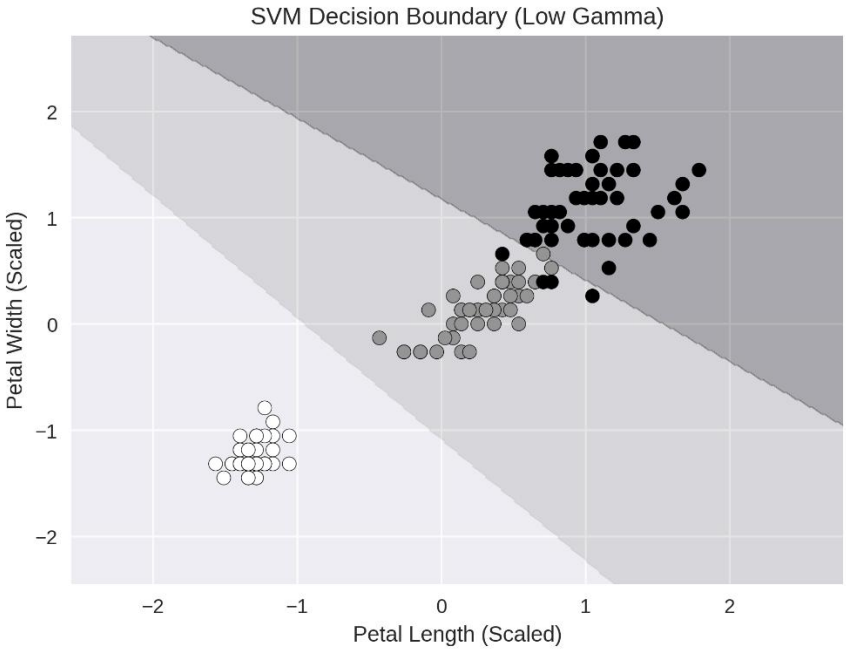


When C is increased, the classifier attempts to classify all training points correctly. As a result, the margin becomes smaller and the decision boundary adapts more closely to the training data.

Influence of the Gamma Parameter

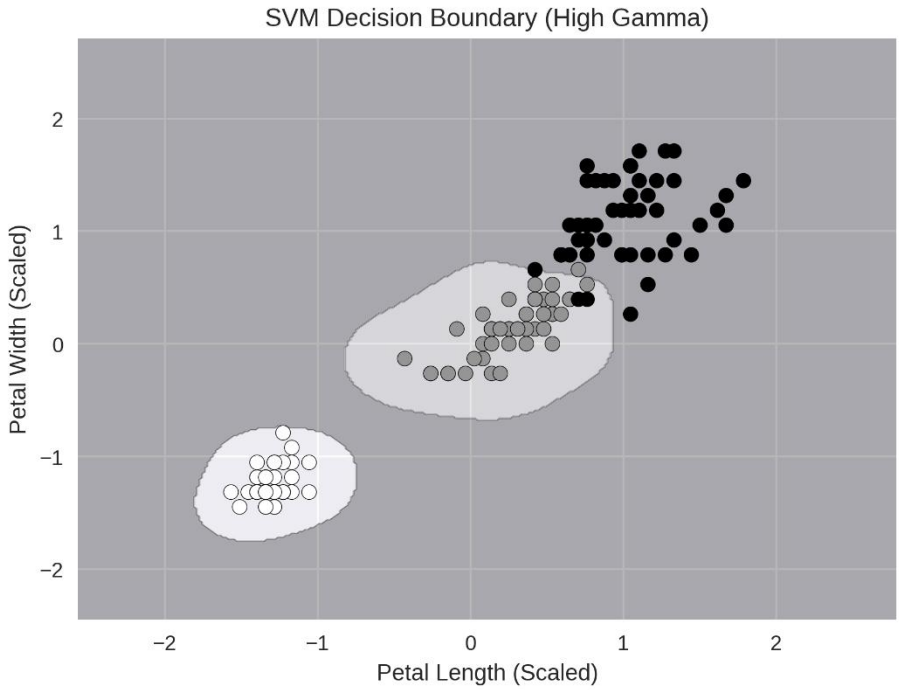
The gamma parameter determines the complexity of the decision boundary.

Decision boundary produced using a low gamma value.



With a low gamma value, the boundary changes gradually across the feature space, resulting in a smooth and generalized classifier.

Decision boundary produced using a high gamma value.

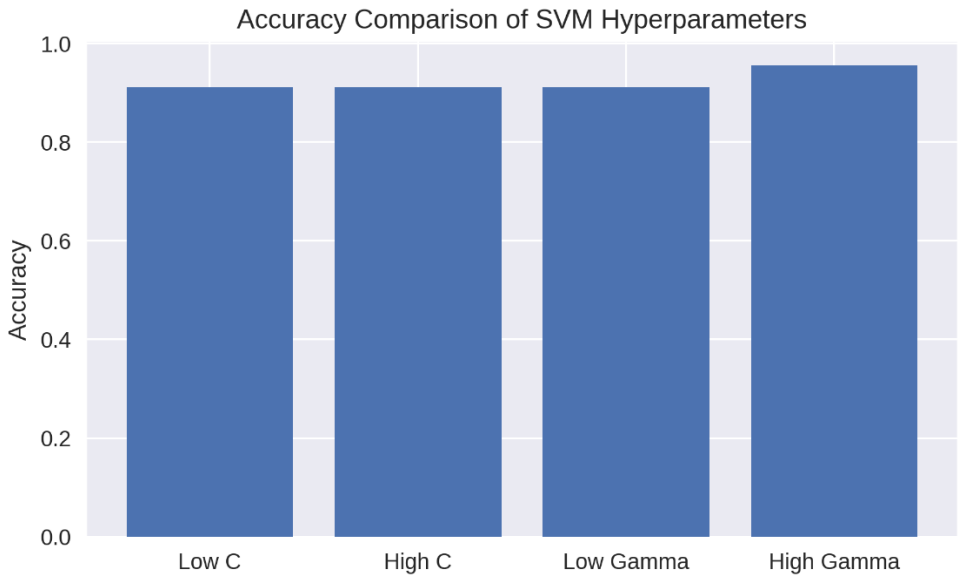


When gamma is large, the classifier becomes highly sensitive to individual observations. This results in a complex boundary that may capture noise in the training data.

Performance Evaluation

To evaluate the effectiveness of the models, several classification metrics were calculated:

- Accuracy -
- Precision
- Recall
- F1 Score



Discussion

According to the results, SVM provide excellent results but we need to tune them carefully.

C parameter affects how much the model tries to correctly classify the training data. With smaller values of C, the margins will be wider and decision boundary will be simple. But larger values will result in more complex decision surface.

The detail of the decision boundary is defined by the gamma parameter. The boundaries become smooth with smaller gamma values, while larger values lead to greater flexibility that could result in overfitting.

The balance between model flexibility and generalization is determined by these parameters.

It is important to choose hyperparameters carefully so that you get accurate predictions.

Limitations:

One limitation of this experiment is that only two features were used for visualization purposes. Although this simplifies the graphical representation of decision boundaries, the full dataset contains additional features that may improve classification accuracy.

Future Work:

Even though this tutorial demonstrates the effect of the C and gamma hyperparameters on Support Vector Machines, there are several extensions that can enhance the analysis and provide insight into their performance.

It is also recommended to experiment with hyperparameter tuning techniques such as Grid Search or Randomized Search. These techniques evaluate different combinations of hyperparameter values to find the one that results in the best model. In the real-world machine learning practice, SVM model tuning optimizes the models as well as avoids underfitting or overfitting.

A potential future extension could be to employ cross-validation, rather than a single train-test split. The cross-validation divides the given data set into multiple training and validation subsets for checking the performance of the model. This method yields a more reliable estimate of how a model performs while minimizing other further risks that may produce biased results.

Conclusion:

This tutorial examined the role of hyperparameter tuning in Support Vector Machines by investigating the influence of the C and gamma parameters.

The experiments demonstrated that these parameters strongly affect both the shape of the decision boundary and the predictive performance of the model.

Understanding how hyperparameters influence model behaviour is essential for building accurate and reliable machine learning systems.

Reference:

Bishop, C.M., 2006. Pattern Recognition and Machine Learning. New York: Springer.

Cortes, C. and Vapnik, V., 1995. Support-vector networks. Machine Learning, 20(3), pp.273–297.

Géron, A., 2022. Hands-On Machine Learning with Scikit-Learn, Keras and TensorFlow. Sebastopol: O'Reilly Media.

Goodfellow, I., Bengio, Y. and Courville, A., 2016. Deep Learning. Cambridge: MIT Press.

Pedregosa, F. et al., 2011. Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research, 12, pp.2825–2830.